

NORMANDIN, QC, Canada

WMO: 713790

Lat: 48.842N

Lon: 72.547W

Elev: 450

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-22.9	-35.8	0.7	-28.0	-30.2	1.0	-22.6	31.4	5.6	28.4	5.8	5.9	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	83.4	67.7	80.0	66.5	76.7	65.1	70.9	79.1	69.0	76.5	67.1	73.9	11.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	105.1	74.9	66.2	98.2	73.1	64.4	92.0	70.9	35.1	79.4	33.5	76.9	31.9	74.0	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	22.3	19.6	-36.0	89.1	5.4	4.0	-39.9	91.9	-43.1	94.2	-46.1	96.5	-50.1	99.3
			-36.4	74.7	5.2	2.1	-40.2	76.2	-43.3	77.5	-46.2	78.7	-50.0	80.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	1.8	4.9	17.3	33.2	49.1	59.1	63.9	61.5	53.4	41.2	27.7	11.6
	DBStd	23.81	14.34	13.81	12.70	8.88	7.91	7.25	5.34	5.63	7.67	7.51	10.18	13.32
	HDD50	6600	1493	1264	1015	506	114	9	0	1	49	290	669	1190
	HDD65	10870	1958	1684	1480	955	495	206	84	139	357	738	1119	1655
	CDD50	1327	0	0	0	1	88	281	432	357	150	17	1	0
	CDD65	119	0	0	0	0	3	27	51	30	8	0	0	0
	CDH74	1322	0	0	0	0	94	362	491	292	80	3	0	0
	CDH80	286	0	0	0	0	21	93	108	52	12	0	0	0
	WSAvg	8.6	9.1	9.2	9.7	9.6	9.0	8.1	7.4	7.2	7.7	8.5	8.8	8.6
(15) Precipitation	PrecAvg	35.9	2.4	1.9	1.9	2.0	3.2	3.2	4.4	4.0	3.7	2.6	2.6	3.1
	PrecMax	44.6	5.7	3.7	4.0	5.5	8.8	6.2	9.0	7.8	6.1	6.1	4.9	5.8
	PrecMin	26.1	0.9	0.3	0.2	0.1	0.4	1.3	2.2	1.7	1.7	0.7	0.5	1.3
	PrecStd	4.5	1.1	1.1	1.0	1.1	1.8	1.3	1.6	1.6	1.2	1.1	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	39.4	45.2	63.1	82.9	87.9	87.1	85.3	81.3	69.6	55.4	38.9
		MCWB	37.5	37.0	39.1	49.6	61.6	68.8	70.0	69.7	68.7	60.0	51.2	36.7
	2%	DB	34.7	34.7	40.6	56.5	75.9	82.2	83.0	80.8	75.7	63.0	48.5	35.8
		MCWB	33.7	32.5	35.8	45.5	59.2	66.1	68.5	68.1	66.2	56.0	45.0	34.4
	5%	DB	30.1	31.6	37.6	50.5	70.1	78.2	79.3	77.2	71.1	57.8	44.1	33.3
		MCWB	28.9	30.2	34.1	42.3	56.4	64.3	67.1	66.2	62.9	52.5	41.6	32.2
	10%	DB	24.3	27.3	35.2	45.7	65.0	73.8	75.9	73.7	67.1	53.2	40.3	31.1
		MCWB	23.2	25.8	32.7	39.3	53.8	61.9	65.5	64.3	60.7	49.2	38.3	30.0
	0.4%	WB	38.0	37.8	40.7	51.7	66.0	72.2	73.6	72.4	70.8	61.6	52.3	37.4
		MCDB	38.9	39.3	44.0	60.4	77.2	82.7	82.4	81.0	79.1	68.1	54.8	38.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.9	33.1	36.6	46.6	61.8	69.2	71.1	69.9	67.7	57.1	45.9	34.5
		MCDB	34.6	34.3	39.4	54.7	69.8	77.7	78.5	77.8	73.9	62.4	48.1	35.6
	5%	WB	28.7	30.0	34.8	43.0	58.0	66.7	69.0	67.9	64.6	53.1	41.7	32.4
		MCDB	29.8	31.6	37.0	49.3	67.7	74.5	76.5	74.8	69.8	56.5	43.6	33.2
	10%	WB	23.0	25.8	32.9	40.0	55.0	64.0	67.0	65.7	61.5	49.8	38.5	29.9
		MCDB	24.1	27.3	35.2	44.7	63.8	71.6	73.7	71.7	66.4	53.0	40.2	31.1
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	22.4	21.6	18.1	22.9	22.6	20.9	21.0	20.3	15.7	13.5	16.7
		MCDBR	19.8	19.2	19.5	25.4	32.3	29.1	25.8	26.2	25.8	23.8	17.2	13.9
		MCWBR	19.3	17.7	16.0	16.3	17.8	15.3	14.1	14.9	16.5	17.1	14.4	13.5
	5% WB	MCDBR	19.5	18.0	17.2	23.6	27.2	24.5	22.6	22.2	23.2	20.8	15.4	14.0
		MCWBR	18.9	16.9	14.6	16.1	16.9	14.5	13.6	13.7	15.9	15.9	13.7	13.7
(40) Clear-Sky Solar Irradiance	taub		0.271	0.285	0.303	0.376	0.355	0.384	0.400	0.377	0.366	0.331	0.322	0.264
	taud		2.229	2.213	2.231	2.227	2.423	2.366	2.342	2.406	2.424	2.498	2.344	2.309
	Ebn at Noon		248	274	288	276	287	278	272	273	265	257	227	238
	Edh at Noon		24	31	36	41	35	37	38	34	30	24	22	19
(44) All-Sky Solar Radiation	RadAvg		418	741	1162	1470	1660	1778	1630	1384	1054	606	324	267
	RadStd		54	94	108	124	151	132	158	88	89	59	39	33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page